Slides 01/12

Méthodes efficaces pour l'estimation de modèles spatio-temporel

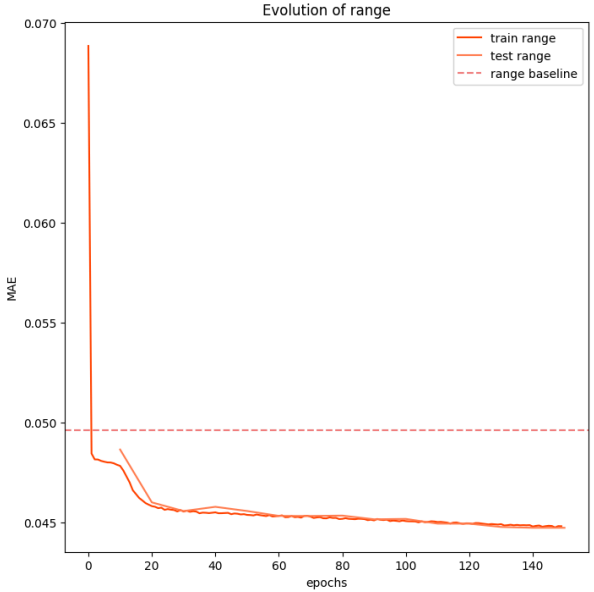
Alexandre Loret

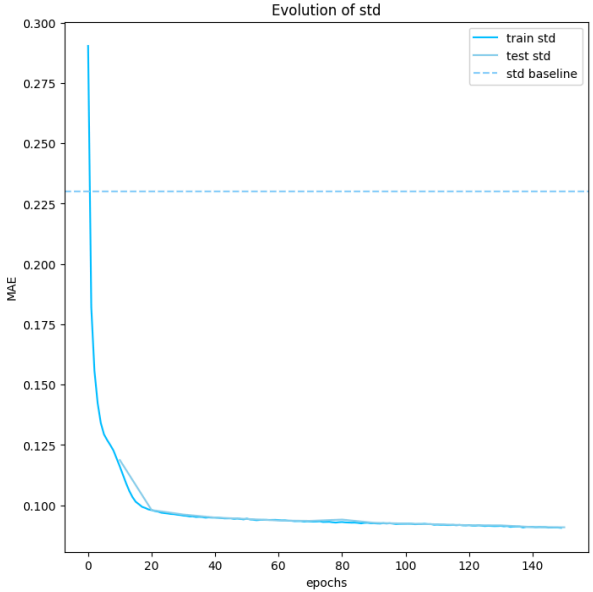
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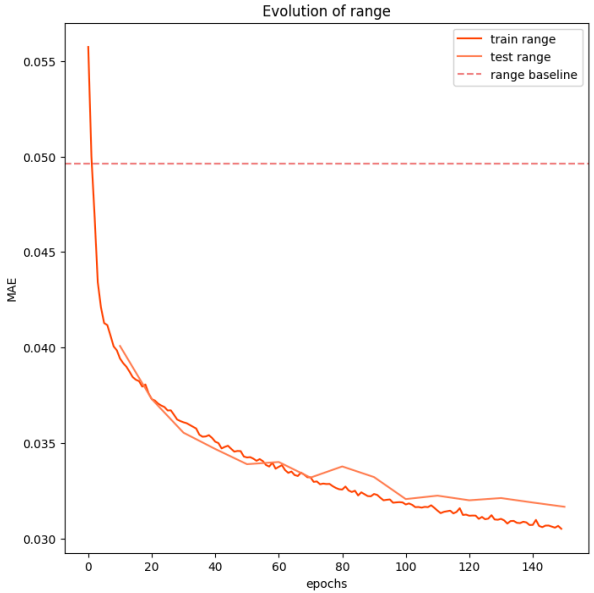
1 Tests plus petits

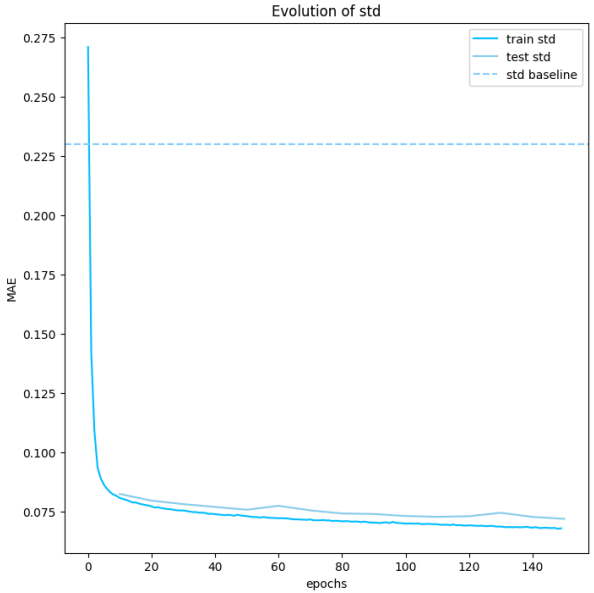
Évolution de la loss



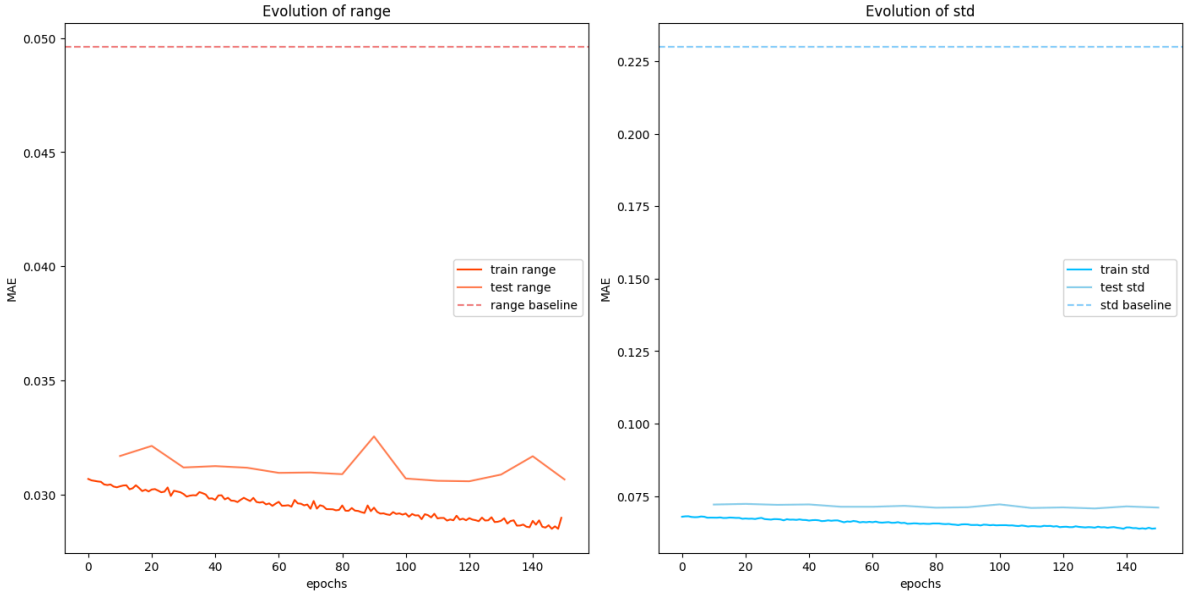


Évolution de la loss (ii)

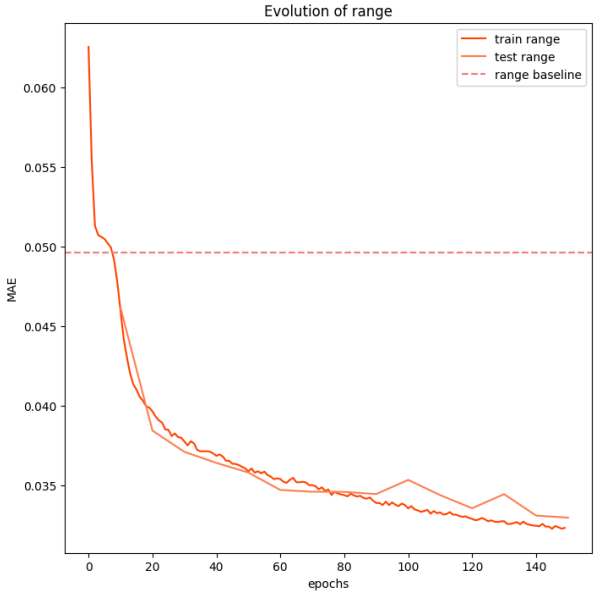


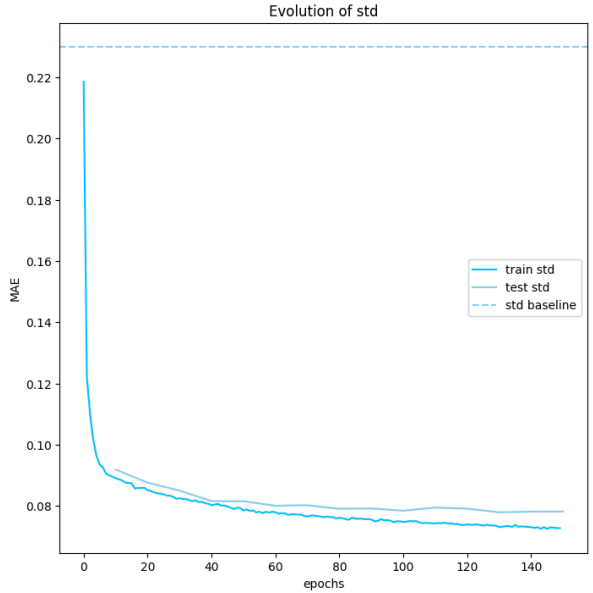


Évolution de la loss (iii)

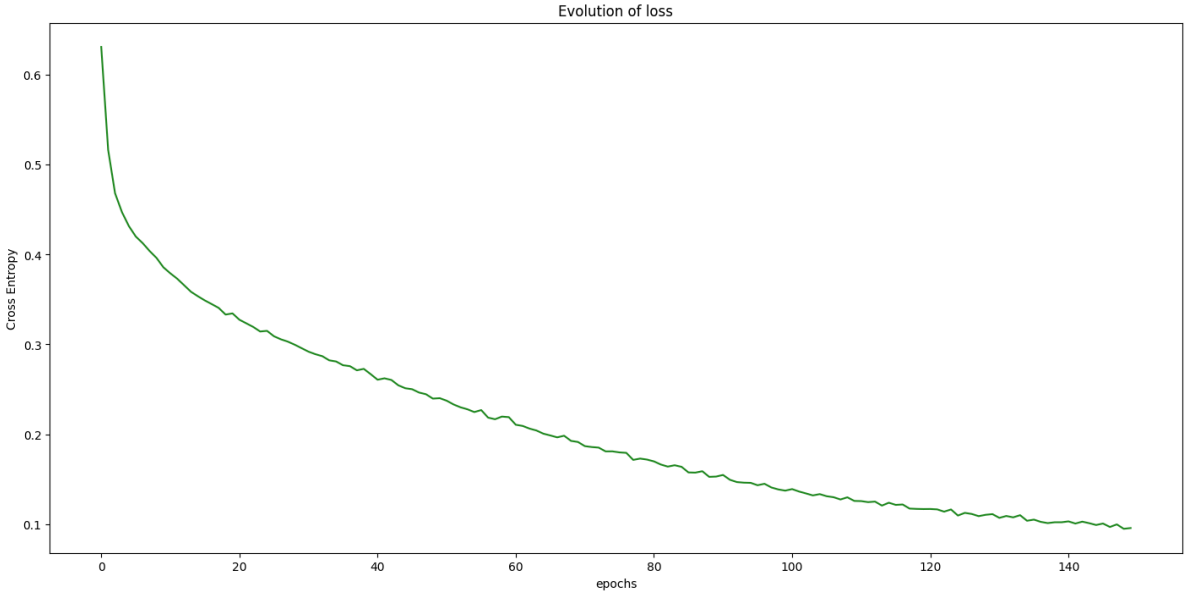


Évolution de la loss (iv)

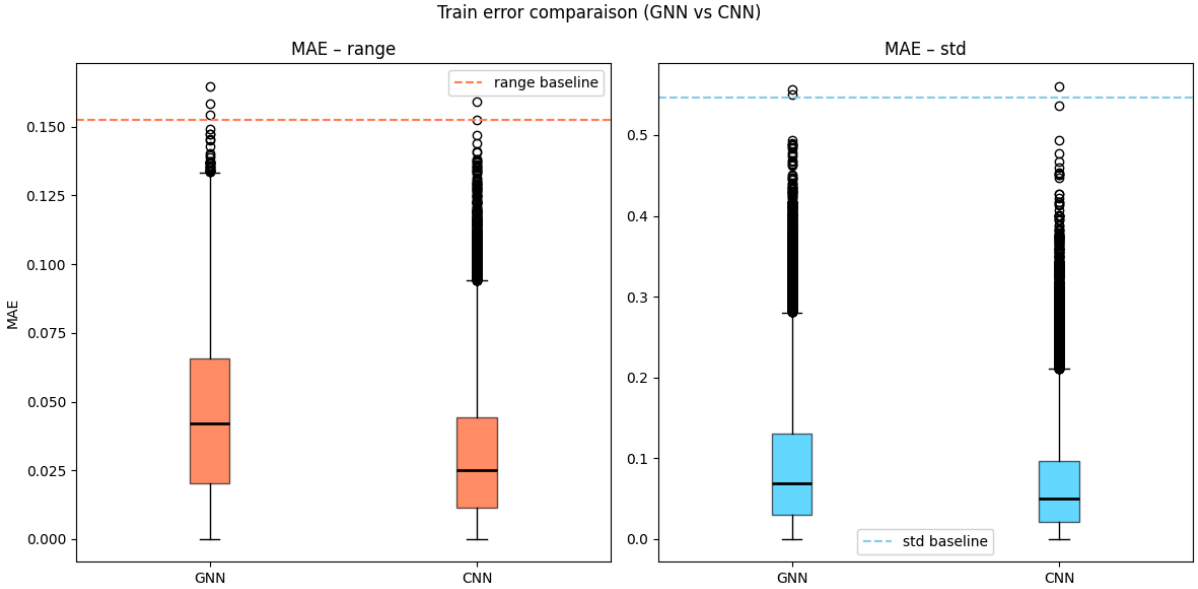




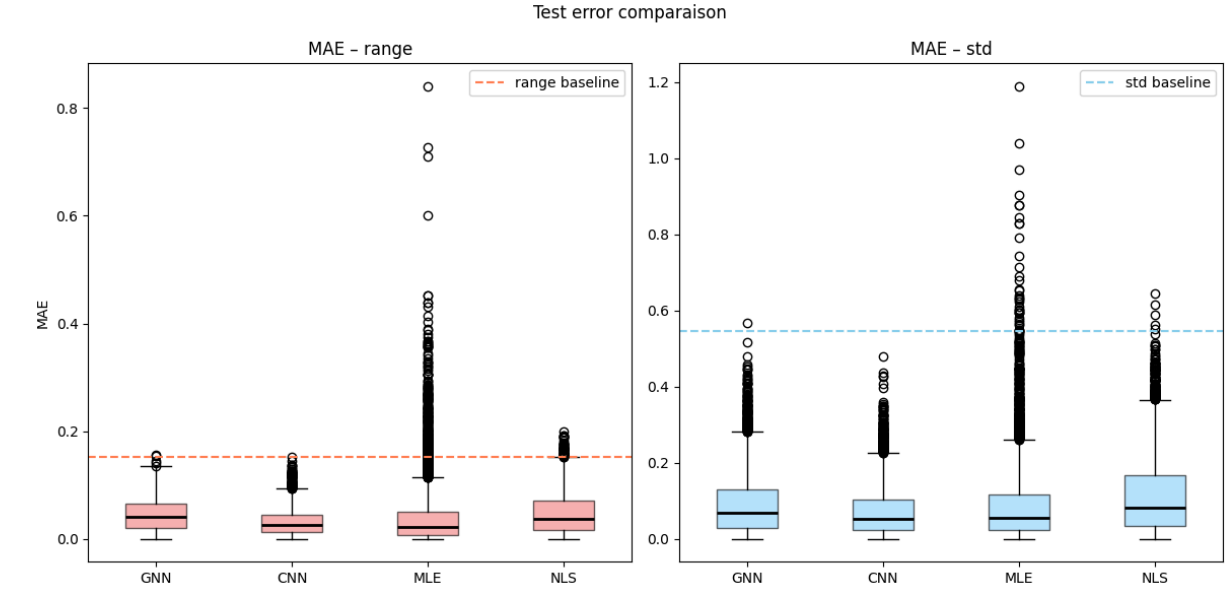
Évolution de la loss (v)



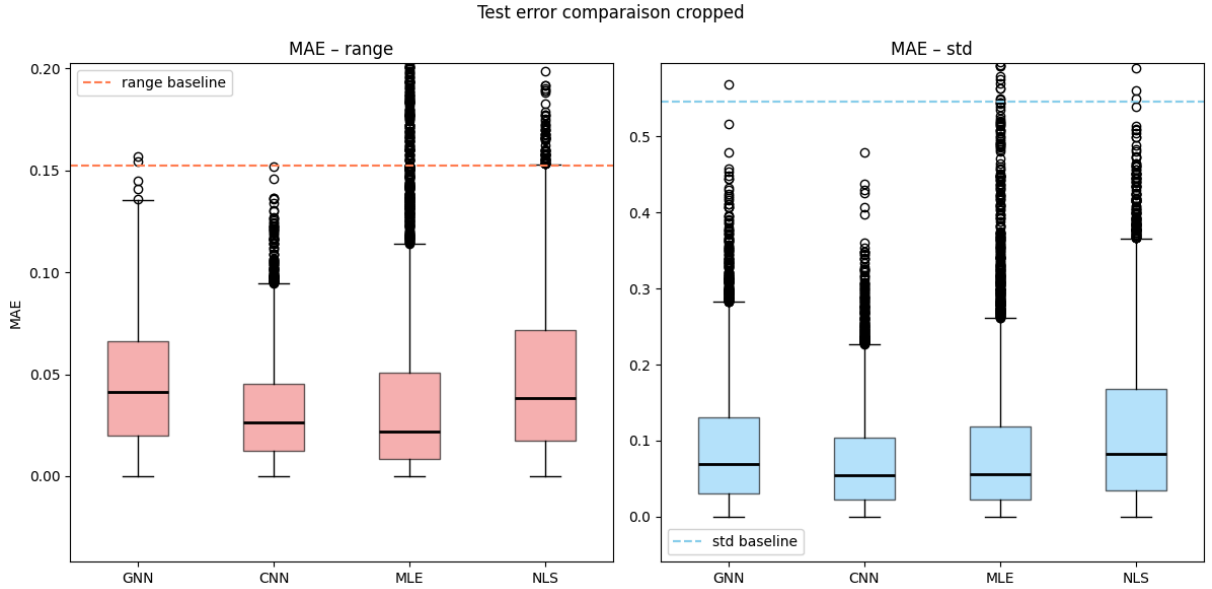
Graphes de conclusion



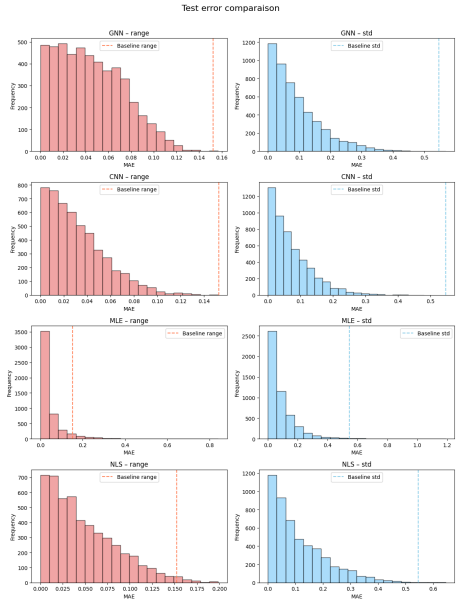
Graphes de conclusion (ii)

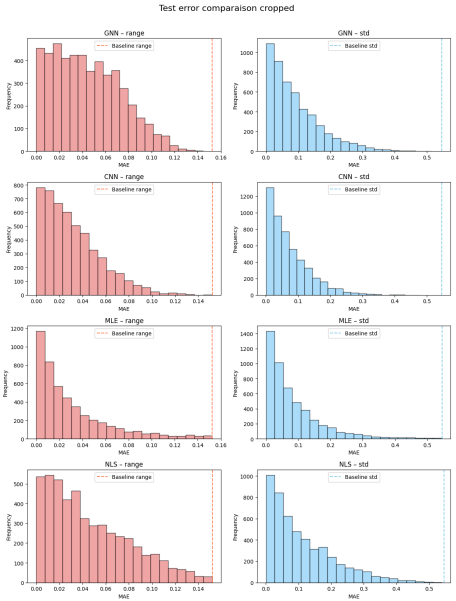


Graphes de conclusion (iii)

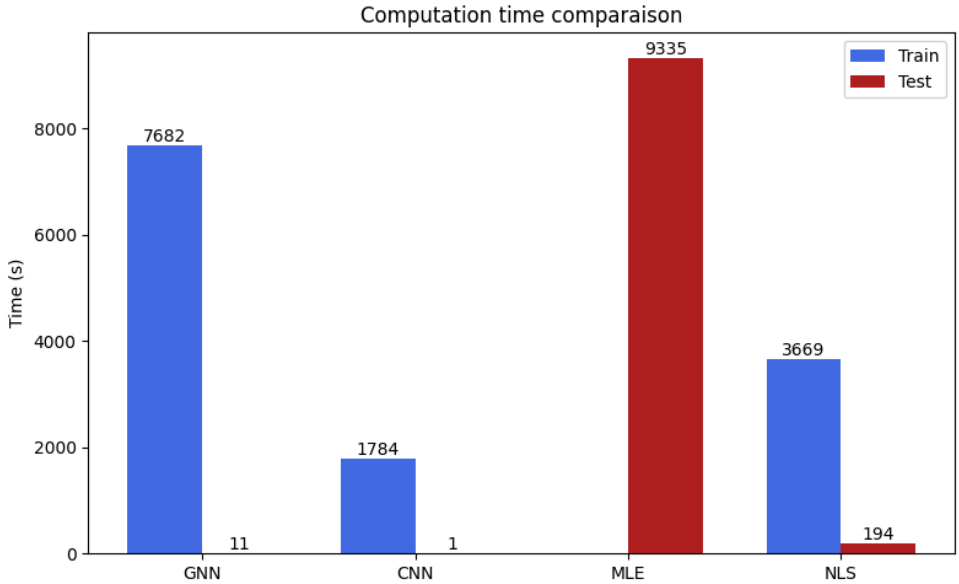


Graphes de conclusion (iv)

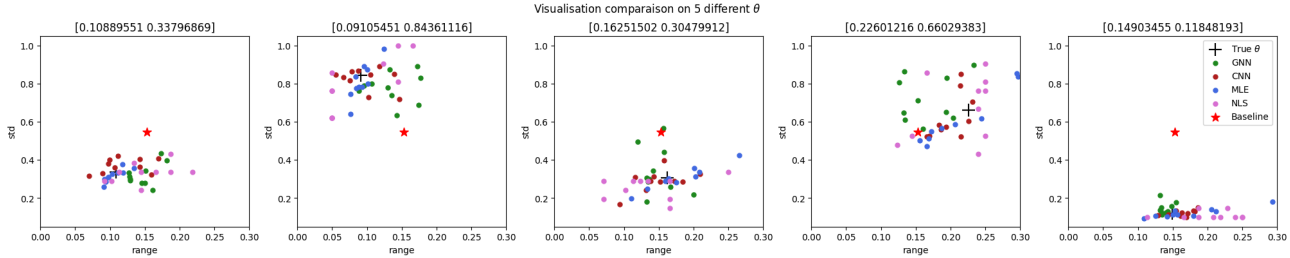




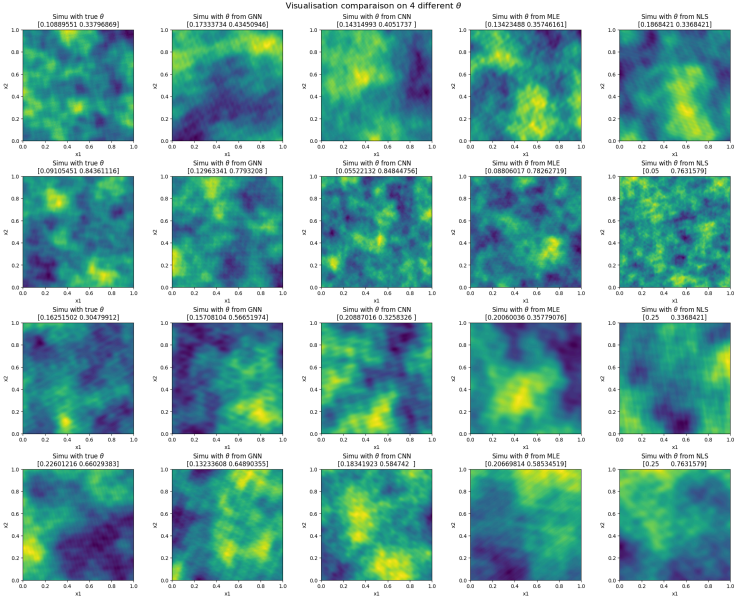
Graphes de conclusion (v)



Graphes de conclusion (vi)



Graphes de conclusion (vii)



2 Nouvelle incertitude

Bootstrap

Donnée entraînement :

$$\begin{pmatrix} (Y_{1,1},\theta_1) & (Y_{1,2},\theta_1) & \dots & (Y_{1,m},\theta_1) \\ (Y_{2,1},\theta_2) & (Y_{2,2},\theta_2) & \dots & (Y_{2,m},\theta_2) \\ & \dots & & \\ (Y_{n,1},\theta_n) & (Y_{n,2},\theta_n) & \dots & (Y_{n,m},\theta_n) \end{pmatrix}$$

Bootstrap (ii)

Une seule observation avec m répliques générées avec θ_{obs} :

$$\begin{pmatrix} Y_{\text{obs},1} & Y_{\text{obs},2} & \cdots & Y_{\text{obs},m} \end{pmatrix}$$

Bootstrap de taille B :

$$\begin{pmatrix} Y_{\text{obs},\pi_1(1)} & Y_{\text{obs},\pi_1(2)} & \cdots & Y_{\text{obs},\pi_1(m)} \\ Y_{\text{obs},\pi_2(1)} & Y_{\text{obs},\pi_2(2)} & \cdots & Y_{\text{obs},\pi_2(m)} \\ & \cdots & & \\ Y_{\text{obs},\pi_{B(1)}} & Y_{\text{obs},\pi_{B(2)}} & \cdots & Y_{\text{obs},\pi_{B(m)}} \end{pmatrix} \overset{\text{modèle}}{\rightarrow} \begin{pmatrix} \theta_1 \\ \theta_2 \\ \cdots \\ \theta_B \end{pmatrix}$$

Chaque π_1, … π_B correspond à m tirages avec remises de 1 à m. On a ensuite un intervalle de confiance avec les quantiles.

Autre méthode

Loss quantile : $L_\tau = \max(\tau * (\text{target} - \text{output}), (1-\tau) * (\text{output} - \text{target}))$

On a trois modèles : $M_{0.5}$, $M_{0.05}$, $M_{0.95}$. La loss est :

$$\begin{aligned} L(Y) = & L_{0.5}(M_{0.5}(Y)) \\ & + L_{0.05}(M_{0.5}(Y) + \exp(M_{0.05}(Y))) \\ & + L_{0.95}(M_{0.5}(Y) + \exp(M_{0.05}(Y)) + \exp(M_{0.95}(Y))) \end{aligned}$$

$M_{0.5}$ donne la médiane (donc MAE) et les deux autres donnent un intervalle de confiance.

3 NLS par minimisation du gradient

Comparaison

Par grid-search :

```
Time to test : 194.94s
Test loss : [0.04823189 0.11234067] ± [0.03774995 0.09952009]
```

Utilisation du package `Pytorch-minimize` avec la méthode `bfgs`

```
Time to test : 114.88s
Test loss : [0.06215605 0.2037503 ] ± [0.04539046 0.13381939]
```

Utilisation du package `Pytorch-minimize` avec la méthode `newton-exact`

```
Time to test : 366.47s
Test loss : [0.06146123 0.18667607] ± [0.04557618 0.1329059 ]
```

4 Spatio-Temporel

Gneiting

Adaptable

Modèles pour grille + masque

Neural Methods for Amortized Inference, Andrew Zammit-Mangion, et al. mentionne des méthodes SBI avec des « Neural Summary Statistics » pouvant être « explice » ou « implicite »

- Implicite :
  - 3D-CNN + Dense
  - 3D-CNN + LSTM (+ Dense)
  - Causal CNN + Dense
- Explicite :
  - **Unsupervised Scalable Representation Learning for Multivariate Time Series, Jean-Yves Franceschi, et al.** (Entraînement non-supervisé par Triplet loss ()
  - **Unsupervised video summarization using deep Non-Local video summarization networks, Sha-Sha Zang, et al.**

Triplet Loss

A faire